

Madhumitha B

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Summary

Final-year B.Tech AI & Data Science student (CGPA 8.5) with 5+ months of industry experience at IITM Research Park. Builds end-to-end AI systems - RAG chatbots, multi-agent systems, computer vision pipelines & RPA automation. IEEE-published researcher and government copyright holder. Targeting full-time AI/ML or Python Developer roles at product-based companies.

Experience

AI Engineer Intern | alfaTKG Co., Ltd. - IITM Research Park

Jan 2026 – Feb 2026

- Built RAG chatbot (FastAPI + Qdrant) with semantic chunking - 87% domain accuracy, ~30% latency reduction; steel defect detection (EfficientNet + PatchCore) achieving 94%+ anomaly accuracy.

AI/ML Intern | Plugzmart, Meras Plugins Pvt. Ltd. - IITM Research Park

Oct 2025 - Jan 2026

- Developed EV charger forecasting model (89% accuracy) for predictive maintenance across 50+ station network; built domain chatbot with real-time charger status & diagnostic data.

Projects

NeuroSphere AI - Multi-Agent RAG System | *FastAPI · Next.js · Groq LLaMA 3.3 70B · Qdrant · MongoDB · LangChain*

- Production-ready ReAct agent (Think→Act→Observe): autonomous intent routing, semantic chunking + MMR reranking, per-user memory, voice I/O (STT/TTS); deployed on Render + Vercel with MongoDB Atlas & Qdrant Cloud.

QueryVault RAG Chatbot | *Python · FastAPI · Qdrant · MongoDB · LangChain*

- End-to-end RAG pipeline - 87% domain accuracy, <1.5s response latency, validated on 200+ real queries.

Steel Surface Defect Detection | *Python · EfficientNet · PatchCore · OpenCV*

- Unsupervised anomaly detection - 94%+ accuracy, 20% false positive reduction via adaptive OpenCV preprocessing.

Dual Hand Sign Language Recognition | *Python · MediaPipe · OpenCV · Flask*

- Real-time dual-hand 26-sign ASL recognition — 91% accuracy, <100ms CPU inference, deployed via Flask UI.

Technical Skills

Languages: Python (Advanced), Java, C, HTML, CSS

AI / ML: TensorFlow, Scikit-learn, YOLOv8, EfficientNet, PatchCore, MediaPipe, OpenCV, Hugging Face

GenAI & NLP: LangChain, RAG Pipelines, Qdrant, FastAPI, Gemini API, Groq, LLaMA 3.3, Prompt Engineering

Tools & DB: UiPath Studio, Docker (basic), GitHub, MySQL, MongoDB, Power BI, Pandas, NumPy

Education

Rajalakshmi Institute of Technology, Chennai

2023 – 2027

B.Tech — Artificial Intelligence & Data Science **CGPA: 8.5 / 10.0**

E.V. Velu MHSS, Tiruvannamalai

2021 – 2023

Higher Secondary (Class XII) **93.4%**

Achievements & Research

- **IEEE Published (2025):** First-authored research paper at IEEE International Conference.
- **Govt. Copyright** SW-39797/2025-CO — AI Web Scraping Chatbot.
- **School Pupil Leader** — coordinated 200+ students.
- **VP, Social Service Club (UBA)** — led 50-member team, 10+ community outreach programs.

Languages

Tamil (Native) · English (Fluent) · German (A1 - Basic)